

my Accounting

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Abstract

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1 Setup python environment

Get a list of python installation:

```
>> type -a python
python is /Users/MANEV/.pyenv/shims/python
python is /usr/bin/python
```

From this list, the first one is used:

```
>> which python
/Users/MANEV/.pyenv/shims/python
```

Check what version of python is in use:

```
>> python -V
Python 3.8.6
```

Check available versions

```
>> pyenv versions
system
3.6.1
* 3.8.6 (set by /Users/MANEV/.pyenv/version)
3.9.16
```

Set proper version in working folder.

```
>> pyenv local 3.9.16
>> python -V
Python 3.9.16
```

Install pipenv to setup our virtual python environment. Notice pipenv doesn't get the version established by pyenv local previously, but the global one. Therefore, you have to care to set it up again in next steps.

```
>> pip install --upgrade pip setuptools wheel
>> pip install pipenv
>> pipenv install --python 3.9.16
>> pipenv install
```

Now you should better move to your working folder. In MacOS you can find the working environment by using the file explorer, in the bottom part, by opening the right mouse button.

```
>> cd working_folder
```

Or by opening an xTerm on the working folder directly, which is directly available on the right side button of the track pad.

Then to launch or create you virtual environment:

```
>> pipenv shell
```

In case you have to remove your virtual environment

```
>> pipenv --rm
```

Install sqlalchemy

```
>> pipenv install sqlalchemy
```

Now you're ready to configure the application for any user.

2 Configuration and administration

There are five tools available for users configuration and for fixing emerging issues in the application:

config_user.py used to create the user's folder, and to copy the generic profile chosen into the specific `<user>_profile.json`. The user profile file declares what accounts are available for use in your accounting, and are moved into `accounts.json` file to create the data base.

showtree.py is able to display accounts as a tree, both in the `accounts.json` file and the database, and to compare them as well.

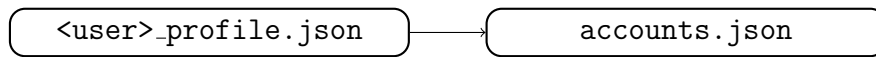
db_tool.py used to create user's data base, and populate it with those accounts declared in the `accounts.json` file, and other useful utilities.

remake.py used to rebuild the user data base and their related yearly closing and opening seats consistently over the years.

cmp_files.py used to compare two seats files. It is helpful to verify changes in the seat files after having saved them with `db_tool save`

2.1 config_user.py

Run `>> python tools/config_user.py <user> create` in order to create your `<user>_profile.json` file, and its subsequent operative application files at due place `\users\<user>\configfiles`:



Thereafter, any modification by hand in your `<user>_profile.json` file, requires you to run `>> python tools/config_user.py <user> refresh` to regenerate consistently the `accounts.json` file.

Commands

```
>> python tools/config_user.py <user> create
>> python tools/config_user.py <user> refresh
```

2.2 db_tool.py

Before starting you have to use this tool to create and prepare the user database. Execute the following sequence:

1. `db_tool.py <user> create` to create user database file
2. `db_tool.py <user> init` to create user database structure
3. `db_tool.py <user> setup` to populate user database with accounts
4. `db_tool.py <user> query` to verify accounts creation
5. `db_tool.py <user> save` to save yearly seats stored in the data base into files named with the tag `app` like `YEAR_app_seats.json`. It's helpful when having to evolve the data base accounts, in companion of the `remake` tool
6. `db_tool.py <user> backup` to create a database file backup
7. `db_tool.py <user> remove` to remove the database file

Commands

```
>> python tools/db_tool.py <user> create
>> python tools/db_tool.py <user> init
>> python tools/db_tool.py <user> setup
>> python tools/db_tool.py <user> query
>> python tools/db_tool.py <user> save
>> python tools/db_tool.py <user> remove
```

2.3 remake.py

It's an important tool that allows to **remake** the user data base and its corresponding accounting yearly closing and opening seat files from the user profile, starting opening seat, and yearly seats file available. It's helpful when having to update the `<user>_opening_seat.json` file, which requires to work it out all data, or when having to modify the `<user>_profile.json` file because new accounts have been created, or its code modified.

Commands

```
>> python tools/remake.py <user>
>> python tools/remake.py <user> -t
>> python tools/remake.py <user> -t new
```

2.4 cmp_files.py

Tool useful to compare json seats files; specifically, `year_<tag>_seats.json` file against `<year>_app_seats.json` file.

Commands

```
>> python tools/cmp_files.py <user> <year>      #tag = <user>
>> python tools/cmp_files.py <user> <year> -t new
```

3 Edition and Report

myAccounting.py is the main operating app for your personal accounting. Seats are edited on its user interface, however since most of the seats are downloaded from the bank account in excel sheets, a specific tool has been made available.

excel.py tool used to take bank accounts excel files to export their seats into json format so that main program can load them.

myReporting.py is available to display data compared to the previous year, and accounts evolution along the years. It's useful when to understand the year.

Commands

```
>> python tools/excel.py <user>
>> python myAccounting.py <user>
>> python myReporting.py <user>
```

3.1 Conceptual Model

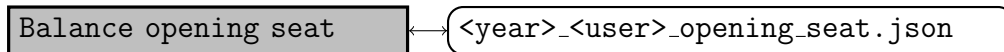
Yet simple, but important to bear in mind:

Seats are held in Journals. BookEntries in Ledgers, and Accounts in the Accounts Book. In data base terms, Journals, Ledgers and Accounts book are tables.

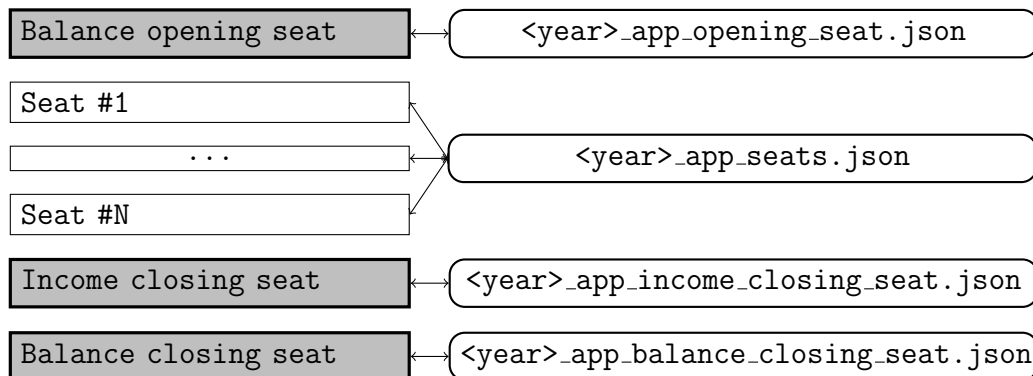
Besides, a seat references several book entries; and each book entry references an account. Therefore, a seat is build by aggregating a number of records in the data base.

Each record holds an identifier that makes it unique. However, when it comes to accounts, besides its unique data base id, they also have a unique code, which we set up in the profile file, and propagates to the `accounts.json` file, and from there to the `accounts` table in the data base.

The first seat is created when starting our accounting. This first seat that holds our opening balance is saved in a file, and from here by using the main app loaded into the data base.



The normal accounting year has three mandatory seats, which are created by the main application upon user request, and are stored in different files. However, they must be loaded by the user to have an effect.



Why are seats stored both in data base and files? The application has been designed to store data both in the data base and in data files for safety reasons mainly. Yet it isn't the only reason. As it is an accounting learning tool, it allows you to evolve your accounting profile, or opening seat, and to **remake** the data base and opening and closing seats consistently.

4 Administration use cases

In order to make operation easy, it's been created a number of tools already introduced at section 2 and a python admin session file, look at subsection 4.6, which opens the database and creates basic variables like db, user, accounts, tree, etc, and some functions which are useful when having to fix issues by hand.

```
>>> list_accounts()
>>> get_account(code)
>>> show_entries_in_account(code)
>>> list_seats()
>>> search_seats_by_description(desc)
>>> get_entry(id)
>>> get_seat(id)
>>> move_entry_to_account(entry_id, code)
>>> move_entries_to_account(entries_list, code)
```

4.1 How to create an user

Run these commands:

```
>> python tools/config_user.py <user> create
>> python tools/db_tool.py <user> create
>> python tools/db_tool.py <user> init
>> python tools/db_tool.py <user> setup
```

then:

```
>> python myAccounting <user>
```

4.2 How to create a new account

When a new account is created, a new entry is created in the Account table. Follow these steps:

1. Edit the `<user>_profile.json` file
 - (a) Open the file, which is located in the folder `configfiles` with a text editor
 - (b) Add a new line, and write the new entry in the **accounts** section


```
{"type": "DEBIT", "content": "NOMINAL", "code": "663", "name": "Tax-House"}
```
 - (c) Save the file
2. Run `>> python tools/config_user.py <user> refresh` to regenerate the application file: `accounts.json`
3. Run `>> python tools/db_tool <user> setup` to update data base account table with the records.

4.3 How to remove an account

To remove an account properly, follow these steps:

1. Clean up the account in the data base by moving its entries to another account, or by removing their seats. Entries belonging to "closing seats" can be left in the account.
2. Save the database `>> python tools/db_tool.py <user> save` into files
3. Edit the `<user>_profile.json` file to remove the target account.
4. Remove the database `>> python tools/db_tool.py <user> remove`
5. Remake it all `>> python tools/remake.py <user>`

4.4 How to modify an account's code

Nominal Account Let's take an example: in the profile we have the following entry:

```
{"type": "DEBIT", "content": "NOMINAL", "code": "621", "name": "ExpH-Water"}
```

where we want to transform the code 621 to become 6211

Follow these steps:

1. Let's make a backup of the data base before modifying it by running `>> python tools/db_tool <user> backup`, that makes a backup copy.
2. Let's edit the `<user>_profile.json` file

- (a) Open the file located in the folder `configfiles` with a text editor.
 - (b) Modify the account's code in the line where it was declared.
`{"type": "DEBIT", "content": "NOMINAL", "code": "6211", "name": "ExpH-Water"}`
 - (c) Save the file.
3. Let's modify account code by following these steps:
- Import session as in 4.6, search the account with code=621 as in ??, then follow:

```
>>> account
Account(33 | Type.DEBIT | Content.NOMINAL | 621 | ExpH-Water | None )
>>> account.code = 6211
>>> mev.db.commit()
>>> quit()
# save db to files
>> python tools/db_tool.py <user> save
# refresh accounts.json
>> python tools/config_user.py <user> refresh
# remake data base from files
python tools/remake.py <user>
```

Real Account The process described for nominal accounts is valid but taking into account a important difference. If the account whose code is going to be modified is used in any of the seats declared in the user opening file: `<user>_opening_seat.json`, there's need to modify this seat before running `>> python tools/remake.py <user>`

4.5 How to modify an account's description

Edit the account description in file `<user>_profile.json`, then follow

```
# update accounts.json
>> python tools/config_user.py <user> refresh

# update db with new info in accounts.json
>> python tools/db_tools.py <user> setup
```

4.6 How to create a python admin session

```
>> pipenv shell
>> python
>>> import admin_session as admin
>>> admin.init('mev') #for mev user
>>> dir(admin)

# if modifying admin_session.py
>>> from importlib import reload

# to reload admin_session.py modified file
>>> reload(admin)

# to commit changes into the database
>>> admin.db.commit()

# to save the database into files
>> python tools/db_tool.py <user> save
```

4.7 Exploring an account

```
# list all accounts available
>>> list_accounts()

# find a specific account description from its code
>>> acc = get_account(code)

# show entries in an account from its code
>>> show_entries_in_account(code, n_items='all')
```

4.8 Finding seats and entries

```
# display seats
>>> list_seats()

# display and get any seat
>>> seat = get_seat(id, prnt=False)

# display and get any entry
>>> entry = get_entry(id, prnt=False)
```

```
# search across seats
>>> search_seats_by_description(desc)
```

4.9 Modify seats and entries

```
>>> seat = get_seat(id)
>>> seat.description = 'Balance Opening Statement'

>>> entry = get_entry(id)
>>> entry.account_id = 70 #moves entry to account 70
# commit database?
```

4.10 Moving entries to another account

```
# move just one entry to another account knowing its code
>>> move_entry_to_account(entry_id, code)

# move a list of entries to another account
>>> move_entries_to_account(ids:list, code)

# commit database?
```

4.11 How to remove an account entry

A Code Snippets

A.1 Exploring an account

Import session as in 4.6, then follow one of the alternatives shown below:

```
#1
>>> acc = next(filter(lambda x: x.code == '64', mev.accounts))
>>> len(acc.entries)
>>> for entry in acc.entries: print(entry)
#2
>>> from dbase import Account, Seat, BookEntry
>>> for item in mev.db.query(Account): print(item)
>>> print(db.query(Account).filter_by(code=645).one())
#3
>>> from sqlalchemy import select
```

```

>>> stmt = select(Account).where(Account.code.in_([64,641]))
>>> for item in mev.db.scalars(stmt): print(item)
#4
>>> stmt = select(BookEntry).join(Account).where(Account.code==622)
>>> print(len(db.scalars(stmt).all()))
>>> for item in db.scalars(stmt): print(item)

```

A.2 How to search seats

Import session as in 4.6 then follow one of the alternatives shown below:

```

#0
>>> admin.seats
#1
>>> seat = next(filter(lambda x:x.id == 3902, admin.seats))
>>> seat
>>> seat.entries
#2
>>> from sqlalchemy import select
>>> from dbase import Seat, BookEntry
>>> stmt = select(Seat).where(Seat.id == 3902)
>>> item = db.scalars(stmt).one()
>>> print(item)
>>> for entry in item.entries: print(entry)
#3
>>> stmt = select(BookEntry).join(Seat).where(Seat.id==3902)
>>> for item in db.scalars(stmt): print(item)
#4
>>> items = db.query(BookEntry).join(Seat).filter_by(id=3902)
>>> for item in items : print(item)

```

A.3 How to search book entries

Import session as in 4.6 then follow one of the alternatives below:

```

#0
>>> mev.entries
#1
>>> entry = next(filter(lambda x:x.id == 4460, mev.entries))
>>> entry.account
>>> entry.Seat

```

```
#2
>>> from sqlalchemy import select
>>> from dbase import BookEntry
>>> for item in db.scalars(select(BookEntry)):print(item)
>>> for item in db.query(BookEntry): print(item)
```